

GraphTheorySymbol Memo

v1.8.0 2026/09/28

Add in your preamble: `\usepackage{graph-theory-symbol}`.

For clarity, below we only show long macro names and short macro names; see Detailed usage section for intermediate length macro names.

Long name	Short name	Result	In context
<code>\GTSadjacency/</code>	<code>\GTSa/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyRelation/</code>	<code>\GTSaR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacency/</code>	<code>\GTSu/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyRelation/</code>	<code>\GTSuR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUp/</code>	<code>\GTSdAU/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpRelation/</code>	<code>\GTSdAUR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUp/</code>	<code>\GTSnDAU/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpRelation/</code>	<code>\GTSnDAUR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDown/</code>	<code>\GTSdAD/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownRelation/</code>	<code>\GTSdADR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDown/</code>	<code>\GTSnDAD/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownRelation/</code>	<code>\GTSnDADR/</code>	$\circ$	$u \circ v$
<code>\GTSincidence/</code>	<code>\GTSi/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceRelation/</code>	<code>\GTSiR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidence/</code>	<code>\GTSnI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceRelation/</code>	<code>\GTSnIR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidence/</code>	<code>\GTSOI/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceRelation/</code>	<code>\GTSOIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidence/</code>	<code>\GTSnOI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceRelation/</code>	<code>\GTSnOIR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidence/</code>	<code>\GTSiI/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceRelation/</code>	<code>\GTSiIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidence/</code>	<code>\GTSnII/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceRelation/</code>	<code>\GTSnIIR/</code>	$\circ$	$v \circ e$

Long name	Short name	Result	In context
\GTSadjacencyMonochrome/	\GTSaM/		u v
\GTSadjacencyMonochromeRelation/	\GTSaMR/		u v
\GTSunadjacencyMonochrome/	\GTSuM/		u v
\GTSunadjacencyMonochromeRelation/	\GTSuMR/		u v
\GTSdirectedAdjacencyUpMonochrome/	\GTSdAUM/		u v
\GTSdirectedAdjacencyUpMonochromeRelation/	\GTSdAUMR/		u v
\GTSnegatedDirectedAdjacencyUpMonochrome/	\GTSnDAUM/		u v
\GTSnegatedDirectedAdjacencyUpMonochromeRelation/	\GTSnDAUMR/		u v
\GTSdirectedAdjacencyDownMonochrome/	\GTSdADM/		u v
\GTSdirectedAdjacencyDownMonochromeRelation/	\GTSdADMR/		u v
\GTSnegatedDirectedAdjacencyDownMonochrome/	\GTSnDADM/		u v
\GTSnegatedDirectedAdjacencyDownMonochromeRelation/	\GTSnDADMR/		u v
\GTSincidenceMonochrome/	\GTSiM/		v e
\GTSincidenceMonochromeRelation/	\GTSiMR/		v e
\GTSnegatedIncidenceMonochrome/	\GTSnIM/		v e
\GTSnegatedIncidenceMonochromeRelation/	\GTSnIMR/		v e
\GTSoutIncidenceMonochrome/	\GTSoIM/		v e
\GTSoutIncidenceMonochromeRelation/	\GTSoIMR/		v e
\GTSnegatedOutIncidenceMonochrome/	\GTSnOIM/		v e
\GTSnegatedOutIncidenceMonochromeRelation/	\GTSnOIMR/		v e
\GTSinIncidenceMonochrome/	\GTSiIM/		v e
\GTSinIncidenceMonochromeRelation/	\GTSiIMR/		v e
\GTSnegatedInIncidenceMonochrome/	\GTSnIIM/		v e
\GTSnegatedInIncidenceMonochromeRelation/	\GTSnIIMR/		v e

\GTSsetMainColor{blue}			
Long name	Short name	Result	In context
\GTSadjacency/	\GTSa/		u v
\GTSadjacencyRelation/	\GTSaR/		u v
\GTSunadjacency/	\GTSu/		u v
\GTSunadjacencyRelation/	\GTSuR/		u v
\GTSdirectedAdjacencyUp/	\GTSdAU/		u v
\GTSdirectedAdjacencyUpRelation/	\GTSdAUR/		u v
\GTSnegatedDirectedAdjacencyUp/	\GTSnDAU/		u v
\GTSnegatedDirectedAdjacencyUpRelation/	\GTSnDAUR/		u v
\GTSdirectedAdjacencyDown/	\GTSdAD/		u v
\GTSdirectedAdjacencyDownRelation/	\GTSdADR/		u v
\GTSnegatedDirectedAdjacencyDown/	\GTSnDAD/		u v
\GTSnegatedDirectedAdjacencyDownRelation/	\GTSnDADR/		u v
\GTSincidence/	\GTSi/		v e
\GTSincidenceRelation/	\GTSiR/		v e
\GTSnegatedIncidence/	\GTSnI/		v e
\GTSnegatedIncidenceRelation/	\GTSnIR/		v e
\GTSoutIncidence/	\GTSoI/		v e
\GTSoutIncidenceRelation/	\GTSoIR/		v e
\GTSnegatedOutIncidence/	\GTSnOI/		v e
\GTSnegatedOutIncidenceRelation/	\GTSnOIR/		v e
\GTSinIncidence/	\GTSiI/		v e
\GTSinIncidenceRelation/	\GTSiIR/		v e
\GTSnegatedInIncidence/	\GTSnII/		v e
\GTSnegatedInIncidenceRelation/	\GTSnIIR/		v e

Long name	Short name	Result	In context
<code>\GTSmetaNot/</code>	<code>\GTSmN/</code>	$\overline{m}$	$\overline{m}^{\circ}$
<code>\GTSmetaAnd/</code>	<code>\GTSmA/</code>	$\wedge^m$	$\wedge^{\circ m}$
<code>\GTSmetaOr/</code>	<code>\GTSmO/</code>	$\vee^m$	$\vee^{\circ m}$

Meta-logic can be used for compound adjacency types, etc.

Implicit meta-and via concatenation, no space except around whole relations compositions, no parentheses via higher priority to meta-and, brackets when all possible adjacencies are specified between two vertices:

Tournament adjacency:

$\forall u, v, u [\overline{m}^{\circ} \vee^{\circ m} v]$.

Tournament adjacency with maybe some colored edge also, who knows?:

$\forall u, v, u [\overline{m}^{\circ} \vee^{\circ m} v]$.

Directed graph:

$\forall u, v, u [\overline{m}^{\circ} \vee^{\circ m} \vee^{\circ m} \vee^{\circ m} v]$.

Oriented graph:

$\forall u, v, u [\overline{m}^{\circ} \vee^{\circ m} \vee^{\circ m} v]$.

Tournament adjacency:

$[\overline{m}^{\circ} \vee^{\circ m}]$ is obtained with `\mathrel{\{[\backslash\text{GTSdAD}/\backslash\text{GTSnDAU}/\backslash\text{GTSmO}/\backslash\text{GTSnDAD}/\backslash\text{GTSdAU}/]\}}`.

Tournament adjacency with maybe some colored edge also, who knows?:

$\overline{m}^{\circ} \vee^{\circ m}$ is obtained with `\mathrel{\{\backslash\text{GTSdAD}/\backslash\text{GTSnDAU}/\backslash\text{GTSmO}/\backslash\text{GTSnDAD}/\backslash\text{GTSdAU}/\}}`.

Directed graph:

$[\overline{m}^{\circ} \vee^{\circ m} \vee^{\circ m} \vee^{\circ m}]$ is obtained with

`\mathrel{\{[\backslash\text{GTSdAD}/\backslash\text{GTSnDAU}/\backslash\text{GTSmO}/\backslash\text{GTSnDAD}/\backslash\text{GTSdAU}/\backslash\text{GTSmO}/\backslash\text{GTSdAD}/\backslash\text{GTSdAU}/\backslash\text{GTSmO}/\backslash\text{GTSnDAD}/\backslash\text{GTSnDAU}/]\}}`.

Oriented graph:

$[\overline{m}^{\circ} \vee^{\circ m} \vee^{\circ m}]$ is obtained with

`\mathrel{\{[\backslash\text{GTSdAD}/\backslash\text{GTSnDAU}/\backslash\text{GTSmO}/\backslash\text{GTSnDAD}/\backslash\text{GTSdAU}/\backslash\text{GTSmO}/\backslash\text{GTSnDAD}/\backslash\text{GTSnDAU}/]\}}`.

Prefix notation:

$\exists u, v, u [\overline{m}^{\circ} \vee^{\circ m} v] \Leftrightarrow \exists u, v, u [\overline{m}^{\circ} \vee^{\circ m} v]$.

$\exists v, e, v [\overline{m}^{\circ} \vee^{\circ m} e] \Leftrightarrow \exists v, e, v [\overline{m}^{\circ} \vee^{\circ m} e]$.

$[\overline{m}^{\circ} \vee^{\circ m}]$ is obtained with

`\mathrel{\{\backslash\text{GTSmO}/\backslash\text{GTSxa}[mC=red,d]\backslash\text{GTSxa}[mC=green,d]\backslash\text{GTSxa}[mC=blue,u]\}}`.

$[\overline{m}^{\circ} \vee^{\circ m}]$ is obtained with

`\mathrel{\{\backslash\text{GTSmO}/\backslash\text{GTSxi}[mC=red,i]\backslash\text{GTSxi}[mC=green,i]\backslash\text{GTSxi}[mC=blue,o]\}}`.

Apart from these symbol macros, versatile macros to call them all are available with options:

`\GTSxa[]`, `\GTSxadj[]`, or `\GTSxadjacency[]` for adjacency symbols,

`\GTSxi[]`, `\GTSxinc[]`, or `\GTSxincidence[]` for incidence symbols,

`\GTSxml[]`, or `\GTSxmetallogic[]` for meta-logic symbols,

`\GTSx[]{}` for all symbols

$(\text{GTSx}[\{a\}] = \text{GTSxa}[], \text{GTSx}[\{i\}] = \text{GTSxi}[], \text{GTSx}[\{m\}] = \text{GTSxml}[])$.

See Detailed usage section for full syntax of these macros.